

DGSSM: Diffusion Guided State-Space Models for Multimodal Salient Object Detection

Suklav Ghosh, Arijit Sur, Pinaki Mitra

Indian Institute of Technology Guwahati, India

Motivation

- **Mamba** provides global receptive fields with **linear computational complexity**, offering an **efficient alternative to quadratic self-attention** for **Salient Object Detection (SOD)**.
- Converting 2D feature maps into 1D sequences can **weaken spatial continuity** during state-space scanning. **Preserving continuous salient regions** is therefore important for complete **object localization and coherent feature representation**.
- Although **Mamba** efficiently captures **global context**, **precise object boundaries and fine-grained structures remain challenging**, a critical requirement for SOD.

Contributions

- ❖ **Diffusion Guided State Space Framework** Reformulates SOD as **progressive denoising**, bridging state-space modeling and diffusion refinement.
- ❖ **Diffusion Structural Prior (DSP)** Injects structural priors into **multi-scale state-space encoding** for **enhanced global coherence**.
- ❖ **Adaptive Saliency Prompting (ASP)** Guides **multi-scale selective scanning** with **saliency-aware prompts**.
- ❖ **Progressive Boundary Refinement** Combines **IMDR** & **BARH** for improved saliency prediction and sharper object contours.

Proposed Model

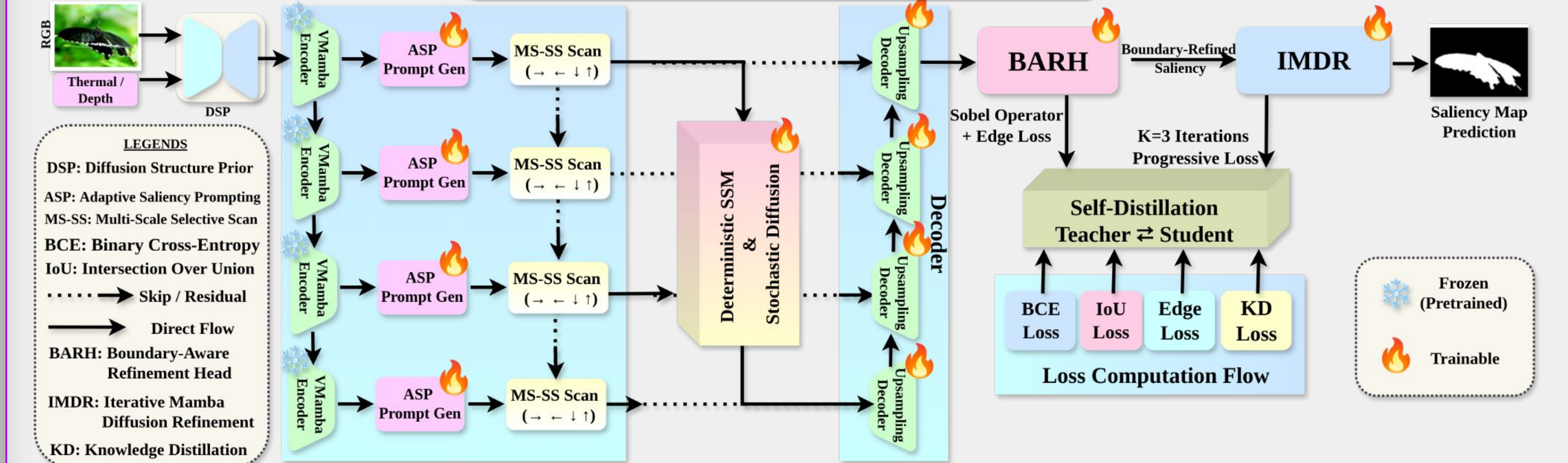


Figure 1: Overall architecture of our proposed Model DGSSM.

Experiments and Results

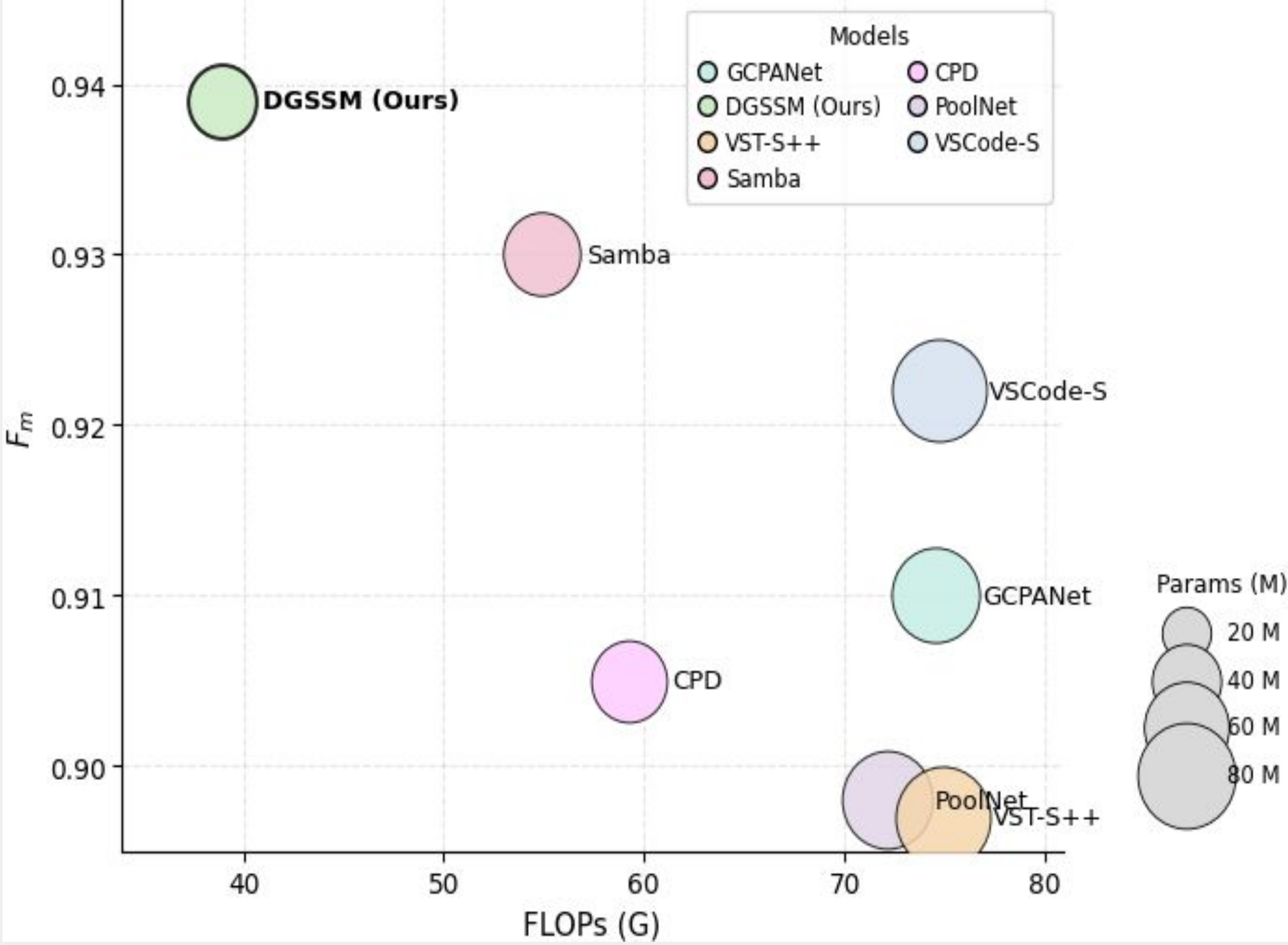


Figure 3: Accuracy efficiency trade-off between FLOPs, performance (F_m), and parameters.

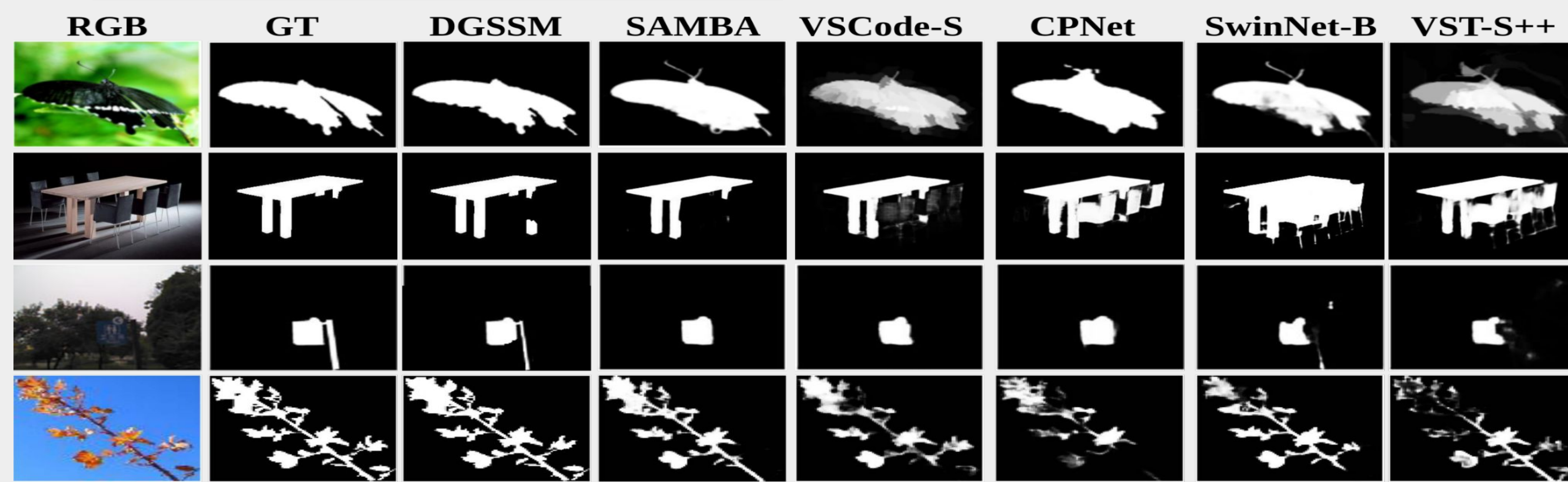


Figure 2: Qualitative Comparison of our DGSSM against SOTA methods

Method	Params (M)	NJUD [51]			NLPR [52]			SIP [53]			STERE [54]			DUTLF-D [55]		
		$S_m \uparrow$	$F_m \uparrow$	$E_m \uparrow$	$S_m \uparrow$	$F_m \uparrow$	$E_m \uparrow$	$S_m \uparrow$	$F_m \uparrow$	$E_m \uparrow$	$S_m \uparrow$	$F_m \uparrow$	$E_m \uparrow$	$S_m \uparrow$	$F_m \uparrow$	$E_m \uparrow$
CNN-based																
JL-DCF [13]	143.52	0.877	0.892	0.941	0.931	0.918	0.965	0.885	0.894	0.931	0.900	0.895	0.942	0.894	0.891	0.927
SP-Net [17]	67.88	0.925	0.928	0.957	0.927	0.919	0.962	0.894	0.904	0.933	0.907	0.906	0.949	0.895	0.899	0.933
SPSN [15]	-	0.925	0.930	0.956	0.918	0.921	0.952	0.923	0.912	0.960	0.892	0.900	0.936	0.907	0.902	0.945
Transformer-based																
SwinNet-B [44]	199.18	0.920	0.924	0.956	0.941	0.936	0.974	0.911	0.927	0.950	0.919	0.918	0.956	0.918	0.920	0.949
CATNet [16]	262.73	0.932	0.937	0.960	0.938	0.934	0.971	0.910	0.928	0.951	0.920	0.922	0.958	0.952	0.958	0.975
VST-S++ [9]	143.15	0.928	0.928	0.957	0.935	0.925	0.964	0.904	0.918	0.946	0.921	0.916	0.954	0.945	0.950	0.969
CPNet [14]	216.50	0.935	0.941	0.963	0.940	0.936	0.971	0.907	0.927	0.946	0.920	0.922	0.960	0.951	0.959	0.974
VSCode-S [42]	74.72	0.944	0.949	0.970	0.941	0.932	0.968	0.924	0.942	0.958	0.931	0.928	0.958	0.960	0.967	0.980
Samba [43]	54.94	0.949	0.956	0.975	0.947	0.941	0.976	0.931	0.949	0.966	0.935	0.933	0.963	0.956	0.964	0.976
DGSSM (Ours)	44.73	0.958	0.964	0.982	0.952	0.950	0.983	0.936	0.959	0.971	0.938	0.937	0.968	0.959	0.967	0.978

Table 1: Comparison of our DGSSM against other SOTA RGB-D SOD methods on five benchmark datasets.

Method	Params (M)	DUTS [46]			DUT-O [47]			HKU-IS [48]			PASCAL-S [49]			ECSSD [50]		
		$S_m \uparrow$	$F_m \uparrow$	$E_m \uparrow$	$S_m \uparrow$	$F_m \uparrow$	$E_m \uparrow$	$S_m \uparrow$	$F_m \uparrow$	$E_m \uparrow$	$S_m \uparrow$	$F_m \uparrow$	$E_m \uparrow$	$S_m \uparrow$	$F_m \uparrow$	$E_m \uparrow$
CNN-based																
CSF-R2 [40]	36.53	0.890	0.869	0.929	0.838	0.775	0.869	-	-	-	0.863	0.839	0.885	0.931	0.942	0.960
EDN [7]	42.85	0.892	0.893	0.933	0.849	0.821	0.884	0.924	0.940	0.963	0.864	0.879	0.907	0.927	0.950	0.957
ICON-R [11]	33.09	0.890	0.876	0.931	0.845	0.799	0.884	0.920	0.931	0.960	0.862	0.844	0.888	0.928	0.943	0.960
MENet [41]	27.83	0.905	0.895	0.943	0.850	0.792	0.879	0.927	0.939	0.965	0.871	0.848	0.892	0.927	0.938	0.956
Transformer-based																
ICON-S [11]	94.30	0.917	0.911	0.960	0.869	0.830	0.906	0.936	0.947	0.974	0.885	0.860	0.903	0.941	0.954	0.971
BBRF [10]	74.40	0.908	0.905	0.951	0.855	0.820	0.898	0.935	0.946	0.936	0.871	0.884	0.925	0.939	0.957	0.972
VST-S++ [9]	74.90	0.909	0.897	0.947	0.859	0.813	0.890	0.932	0.941	0.969	0.880	0.859	0.901	0.939	0.951	0.969
VSCode-S [42]	74.72	0.926	0.922	0.960	0.877	0.840	0.912	0.940	0.951	0.974	0.887	0.864	0.904	0.949	0.959	0.974
Samba [43]	49.59	0.932	0.930	0.966	0.889	0.859	0.922	0.945	0.956	0.978	0.892	0.896	0.931	0.953	0.965	0.978
DGSSM (Ours)	38.92	0.947	0.939	0.978	0.903	0.869	0.930	0.951	0.959	0.981	0.899	0.901	0.934	0.963	0.973	0.983

Table 2: Comparison of our DGSSM against other SOTA RGB-T SOD methods on three benchmark datasets.

Conclusion

- ★ **Mamba** enables **long-range reasoning** with **linear complexity**.
- ★ **DSP and ASP** modules improve **global coherence** and **spatial representation**.
- ★ **IMDR & BARH** modules deliver **sharper object contours**.
- ★ **DGSSM** achieves **consistent SOTA performance across 13 RGB, RGB-D, RGB-T benchmarks** with a compact model footprint.
- ★ Achieves superior accuracy with **lower FLOPs and fewer parameters**.
- ★ Generalizes across **RGB, RGB-D and RGB-T** modalities.
- ★ **Efficient and scalable** for high-resolution dense prediction.

DGSSM synergistically combines the efficiency of state space models and the structural power of diffusion models, setting a new benchmark for salient object detection.

References

- J. He, K. Fu, X. Liu, and Q. Zhao. Samba: A unified Mamba-based framework for general salient object detection. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pages 25314–25324, 2025.
- Luo, Z., et al.: VSCode: General visual salient and camouflaged object detection with 2D prompt learning. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 17169–17180, 2024.
- Yue Liu, Yunjie T., Zhao Y., et al.: VMamba: visual state space model. In Proceedings of the 38th International Conference on Neural Information Processing Systems (NeurIPS), Vol. 37, 2024.
- Dong, W., et al.: Fusion-Mamba for cross-modality object detection. IEEE Transactions on Multimedia, 2025.

